

Aptitude — Quick Reference Notes

PlacementPrep Study Notes

1. Time, Speed & Distance

Core relation: $\text{Speed} = \text{Distance} / \text{Time}$.

- Convert km/hr to m/s by multiplying by $5/18$, and m/s to km/hr by multiplying by $18/5$.
- For a train crossing a pole, distance = length of train.
- For a train crossing a platform, distance = length of train + length of platform.
- Average speed for equal distances at speeds a and $b = 2ab / (a + b)$.

2. Time & Work

Work done is inversely proportional to time taken.

- If A can do a job in ' x ' days, A's one-day work = $1/x$.
- Combined one-day work of A and B = $1/x + 1/y$.
- Total time for both working together = $1 / (\text{combined one-day work})$.

3. Percentages, Profit & Loss

Percentage change = $(\text{Change} / \text{Original value}) \times 100$.

- Profit % = $(\text{Profit} / \text{Cost Price}) \times 100$.
- Loss % = $(\text{Loss} / \text{Cost Price}) \times 100$.
- Selling Price = $\text{Cost Price} \times (1 + \text{Profit\%} / 100)$.

4. Simple & Compound Interest

$\text{SI} = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$.

- Amount in Compound Interest = $P \times (1 + R/100)^T$.
- For the same rate and time, Compound Interest is always greater than or equal to Simple Interest.

5. Ratio & Proportion, Averages

Ratio compares two quantities of the same kind.

- If $a:b = m:n$, then $a/b = m/n$.
- Average = $\text{Sum of observations} / \text{Number of observations}$.

Practice Tip

- Time yourself: aim for under 60 seconds per aptitude question in a mock test.
- Memorize squares up to 30 and cubes up to 15 to speed up calculations.